

IoT, Cyber-Physical Systems and Security

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The lecture maps to the following CyBOK Knowledge Areas:

Full Coverage:

- **Infrastructure Security → Cyber-Physical Systems Security**

Partly Covered:

- **Infrastructure Security → Network Security**
- **Systems Security → Authentication, Authorisation & Accountability**
- **Human, Organisational & Regulatory Aspects → Risk Management & Governance**

INTRODUCTION TO IoT AND CPS

IoT AND CPS ARCHITECTURE

SECURITY FOUNDATIONS

REAL-WORLD EXAMPLES

Introduction to IoT and CPS

WHAT IS IoT?

The **Internet of Things (IoT)** is a network of physical devices connected to the Internet that can collect, exchange, and process data.

IoT technologies are used in a wide range of environments to support monitoring, automation, and data-driven decision making.

IoT technologies are used across many sectors to monitor physical environments, exchange data, and support automated decision making. Examples of common application domains include:

- Consumer IoT (smart homes, wearables)
- Healthcare IoT (medical monitoring, connected devices)
- Industrial IoT (manufacturing, process control)
- Transportation (connected and autonomous vehicles)
- Smart Cities (traffic, lighting, public services)
- Environmental and Agricultural Monitoring

WHAT IS A CYBER-PHYSICAL SYSTEM?

A **Cyber-Physical System (CPS)** integrates computation, communication, and physical processes to monitor and interact with the physical world.

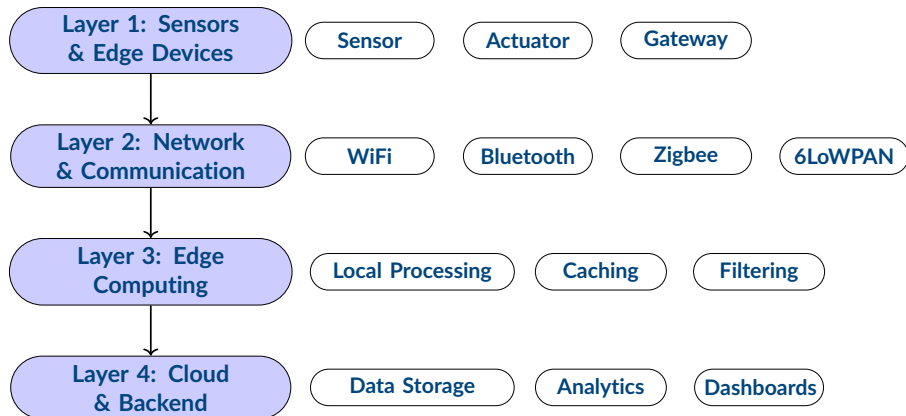
- Sensors collect information from the physical environment.
- Networks transfer data between system components.
- Software analyses data and makes decisions.
- Actuators perform actions that affect the physical world.

Cyber-Physical Systems are used in environments where software interacts with and influences physical processes. Examples of common application domains include:

- Industrial Control Systems (manufacturing, process control)
- Healthcare Systems (medical devices, patient monitoring)
- Transportation Systems (connected and autonomous vehicles)
- Smart Grids (energy generation and distribution)
- Smart Buildings (HVAC, lighting, access control)
- Critical Infrastructure (water treatment, rail, power systems)

IoT and CPS Architecture

IoT REFERENCE ARCHITECTURE



Data flows up the stack; control flows down. Each layer has different **security requirements**.

■ Short-Range and Low-Power Wireless

- **Bluetooth / BLE:** Wearables and personal devices
- **Zigbee:** Home automation and mesh networks
- **6LoWPAN:** IPv6 over constrained low-power wireless networks
- **NFC:** Contactless communication

■ Medium-Range Wireless

- **WiFi:** Home and enterprise networks

■ Long-Range Wireless

- **Cellular (LTE/5G):** Mobile and remote deployments
- **LoRaWAN:** Low-power wide-area networks

Messaging protocols define how IoT data and commands are exchanged above the underlying network technology.

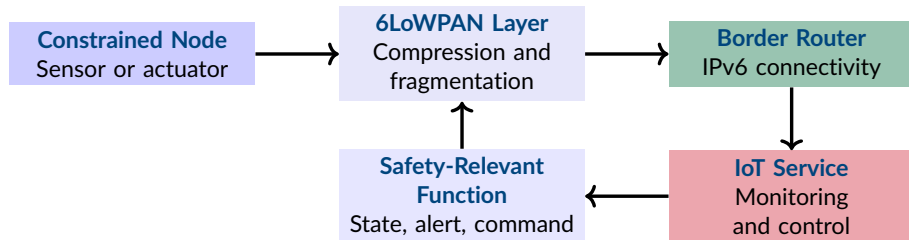
- **MQTT:** Lightweight publish-subscribe messaging commonly used by sensors, gateways and dashboards
- **CoAP:** Lightweight request-response protocol designed for constrained devices
- **HTTP/REST:** Common web-based communication model, but usually heavier than MQTT or CoAP

6LoWPAN

Enables **IPv6 communication** over constrained low-power wireless networks, such as IEEE 802.15.4.

- **Purpose:** Connect constrained sensors and actuators to IP-based networks
- **Constraint:** Devices often have limited power, memory, processing and frame size
- **Adaptation Layer:** Supports header compression, fragmentation and reassembly
- **Border Router:** Connects the low-power wireless network to wider IPv6 networks
- **Safety-Relevant Use:** May carry readings, commands, alerts or device state

6LoWPAN ARCHITECTURE



6LoWPAN brings **IP networking** into constrained wireless systems that may influence physical behaviour.

CYBER-PHYSICAL SYSTEM (CPS) ARCHITECTURE



Feedback loop: cyber decisions influence the physical process, and physical changes generate new data.

Key Characteristics of Cyber-Physical Systems

- CPS combine cyber components with physical processes.
- Sensors continuously monitor the physical environment.
- Decisions made by software can affect the physical world.
- Failures may have safety, operational, or financial consequences.

- **Physical Process:** The real-world system being monitored or controlled.
- **Sensors:** Collect information from the physical environment.
- **Communication Network:** Transfers data between system components.
- **Controllers:** Analyse data and make decisions.
- **Actuators:** Perform actions that influence the physical process.
- **Feedback Loop:** Continuous interaction between cyber and physical components.

Security Foundations

■ Confidentiality

- Protecting sensitive information from unauthorised disclosure

■ Integrity

- Ensuring data and system behaviour remain trustworthy

■ Availability

- Maintaining access to devices, services, and communications

■ Authenticity

- Verifying the identity of devices and users

■ Safety

- Preventing security failures from causing physical harm

Why is IoT security different from traditional IT?

■ Resource Constraints

- Limited computing power, memory, and battery life

■ Diversity

- Large variety of devices, vendors, and technologies

■ Lifecycle

- Long deployment periods with infrequent updates

■ Connectivity

- Continuous communication over wired and wireless networks

■ Physical Exposure

- Devices are often deployed in accessible environments

■ Safety Impact

- Security failures may affect physical processes and safety

What threats can affect connected devices and physical processes?

■ Eavesdropping

- Intercepting communications to obtain sensitive information

■ Data Manipulation

- Modifying sensor readings or control commands

■ Device Impersonation

- Spoofing a trusted device or service

■ Denial of Service

- Disrupting communications or system availability

■ Physical Tampering

- Gaining physical access to devices, sensors, or actuators

How can IoT systems be attacked?

■ Device Layer

- Physical access, firmware extraction, key theft

■ Network Layer

- Interception, disruption, or manipulation of communications

■ Cloud and Applications

- Compromise of services, databases, or user accounts

■ Supply Chain

- Vulnerabilities introduced during development or updates

■ Ecosystem

- Compromise of gateways or interconnected devices

How can we protect IoT devices and cyber-physical systems?

■ Authentication

- Verify the identity of users, devices, and services

■ Access Control

- Restrict access to authorised entities

■ Encryption

- Protect data during storage and transmission

■ Secure Updates

- Patch vulnerabilities throughout the device lifecycle

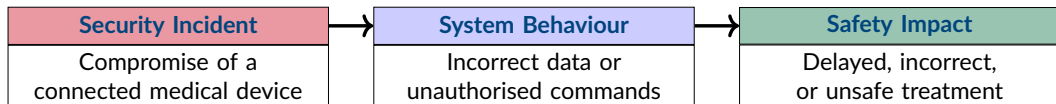
■ Monitoring

- Detect abnormal behaviour and potential attacks

Real-World Examples

REAL-WORLD EXAMPLE: HEALTHCARE

Security failures can become patient safety issues

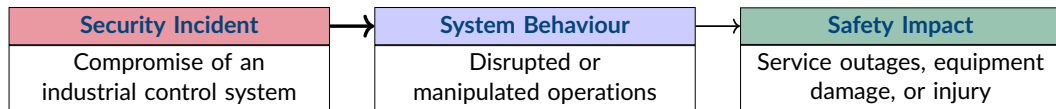


Example

Researchers have demonstrated attacks against connected insulin pumps and pacemakers.

REAL-WORLD EXAMPLE: CRITICAL INFRASTRUCTURE

Cyber attacks can affect physical infrastructure



Example

Recent incidents have targeted pipelines, power systems, and water treatment facilities.

- **Internet of Things (IoT)** connects physical devices, sensors, and services through communication networks
- **Cyber-Physical Systems (CPS)** integrate computation, communication, and physical processes
- **IoT and CPS architectures** consist of multiple layers from devices to cloud services
- **Communication technologies and protocols** enable data exchange between system components
- **Security objectives** include confidentiality, integrity, availability, authentication, and privacy
- **IoT systems face unique challenges** due to resource constraints, diversity, connectivity, and lifecycle issues
- **Security incidents** can influence system behaviour and lead to real-world safety consequences

- NCSC. *Device security principles for manufacturers*. [\[Link\]](#)
- NIST. (2019). *Cyber-Physical Systems and Internet of Things*. NIST SP 1900-202. [\[Link\]](#)
- ENISA. (2020). *Guidelines for Securing the Internet of Things*. [\[Link\]](#)
- OWASP. *Internet of Things Project (Top 10)*. [\[Link\]](#)